

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 January 2026
Team ID	LTVIP2026TMIDS38297
Project Name	Gemini Historical Artifact Description
Maximum Marks	4 Marks

Technical Architecture:

Table-1: Components & Technologies

S.No	Component	Description	Technology
1.	User Interface	Museum-themed web interface for users to input text or upload images.	Start: Streamlit (Python-based UI framework), HTML, CSS (Custom Styling)
2.	Application Logic	Core logic for handling user inputs, API communication, and state management.	Python 3.8+
3.	Generative AI Service	AI Engine for generating historical descriptions and analyzing images.	Google Gemini 2.5 Flash
4.	Image Processing	Logic for handling and encoding uploaded images before API transmission.	Pillow (PIL) Library
5.	Environment Config	Management of sensitive credentials like API keys.	python-dotenv
6.	File Storage	Temporary in-memory handling for uploads; local storage for downloads.	Python io module , Local Filesystem

7.	External API	The primary AI service used for content generation.	Google Generative AI API
8.	Infrastructure	Deployment environment for running the Python application.	Local Machine (Development) / Streamlit Community Cloud (Production)

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	The core framework used to build and deploy the web application quickly.	Streamlit
2.	Security Implementations	Secure handling of API keys to prevent exposure.	Environment Variables (.env) / Streamlit Secrets
3.	Scalable Architecture	The app is stateless, allowing multiple instances to run independently behind a load balancer.	Cloud-Native / Serverless (when deployed on Cloud)
4.	Availability	High availability is managed by the cloud platform hosting the app.	Streamlit Cloud / Docker Container orchestration
5.	Performance	Basic caching of static resources and optimized API calls for faster response.	Streamlit Caching (<code>@st.cache_data</code>), Gemini Flash Model (Low Latency)